

Prithwish Dan

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EDUCATION

Cornell University

Ithaca, NY

Master of Science in Computer Science, GPA - 4.250

Expected Graduation: May 2026

- Advisors: Sanjiban Choudhury, Wei-Chiu Ma - ML, Robotics, Imitation Learning, Reinforcement Learning, Sim-to-Real

Cornell University

Ithaca, NY

Bachelor of Science in Computer Science, GPA - 4.135 (Summa Cum Laude)

Aug. 2020 - May 2024

- Coursework: Computer Vision, Robot Learning, Large-Scale Machine Learning, Operating Systems, Analysis of Algorithms

PUBLICATIONS

1) X-Sim: Cross-Embodiment Learning via Real-to-Sim-to-Real

CoRL 2025

Prithwish Dan*, Kushal Kedia*, et al., Wei-Chiu Ma, and Sanjiban Choudhury

Accepted (Oral)

- Designed framework to generate synthetic robot data from human videos using object-centric RL with zero tele-operation
- Introduced novel sim-to-real adaptation scheme to automatically minimize the visual gap at deployment time
- Received Best Paper Finalist award @ EgoAct: Workshop on Egocentric Perception and Action for Robot Learning RSS 2025

1) One-Shot Imitation under Mismatched Execution

ICRA 2025

Kushal Kedia*, Prithwish Dan*, et al., and Sanjiban Choudhury

Accepted

- Developed novel framework to align visual representations for cross-embodiment imitation learning using optimal transport
- Outperformed prior works by 50%+ in task completion rate across datasets with different levels of embodiment mismatch

2) MOSAIC: A Modular System for Assistive and Interactive Cooking

CoRL 2024

Yuki Wang, Kushal Kedia, Juntao Ren, Prithwish Dan, et al., and Sanjiban Choudhury

Accepted

- Developed a modular architecture for coordinating multiple robots to interact and collaborate with users in the kitchen
- Won best paper award @ VLNMN WORKSHOP and best poster award @ MOMA WORKSHOP at ICRA 2024

3) InteRAct: Transformer Models for Human Intent Prediction Conditioned on Robot Actions

ICRA 2024

Kushal Kedia, Atiksh Bhardwaj, Prithwish Dan, and Sanjiban Choudhury

Accepted

- Implemented novel algorithm to predict human intent conditioned on robot actions for collaborative manipulation
- Reduced forecasting errors by 2x for 3 human-robot tasks by pre-training on large-scale human-human activity data

4) ManiCast: Collaborative Manipulation with Cost-Aware Human Forecasting

CoRL 2023

Kushal Kedia, Prithwish Dan, Atiksh Bhardwaj, and Sanjiban Choudhury

Accepted

- Formulated new learning objective to learn cost-aware human motion forecasts for human-robot interactions
- Improved downstream planning metrics such as reaction time and time-to-goal by over 25% relative to baselines

5) A Game-Theoretic Framework for Joint Forecasting and Planning

IROS 2023

Kushal Kedia, Prithwish Dan, and Sanjiban Choudhury

Accepted

- Proposed method to jointly train a forecaster and planner to encourage safer autonomous planning in crowd settings

EXPERIENCE

Apple

Sunnyvale, CA

Machine Learning Research Intern

May 2025 – Present

- Building multi-modal (IMU, audio) foundation models for resource-constrained wearable devices to support downstream apps
- Researching distillation from large-scale teacher models and investigating power and performance tradeoffs

PoRTaL Lab

Ithaca, NY

Machine Learning/Robotics Researcher

Jan. 2023 – Present

- Leading research initiatives to leverage human videos and foundation models to scale up data generation for robot learning
- Building novel human intent prediction frameworks in PyTorch and ROS to enable seamless human-robot collaboration

AWARDS

Merrill Presidential Scholar 2024

[Webpage](#)

- Honors Top 1% of all Cornell undergraduates in academic achievement and leadership

CRA Outstanding Undergraduate Researcher 2024 - Honorable Mention

[Webpage](#)

- Recognizes top undergraduates across North America with outstanding research potential in a computational field

TECHNICAL SKILLS

Languages: Python, C/C++, Java, OCaml, JavaScript/TypeScript

Frameworks/Technologies: PyTorch, OpenCV, NumPy, Tensorboard, Pandas, ROS, JUnit, Flask, React, Bazel

Developer Tools: Git, Linux/Unix, Windows, VSCode, JetBrains, Postman

ADDITIONAL EXPERIENCE

MongoDB - SF

Software Engineering Intern

San Francisco, CA

June 2024 – Aug. 2024

- Developed novel APIs using Java backend to enable secure Cloud data encryption via Azure private networking

MongoDB - NYC

Software Engineering Intern

New York City, NY

June 2023 – Aug. 2023

- Implemented novel slot-based query execution algorithm in C++ for MongoDB's full-text search feature, achieving 40%+ increased speeds and 10x reduced memory usage, providing users with improved database experiences
- Collaborated cross-functionally with core server and cloud teams to ensure efficient data transfer in query pipelines

Cornell Bowers CIS

Teaching Assistant

Ithaca, NY

Jan. 2022 – Present

- Holding office hours, review sessions, grading exams/assignments, and leading discussion sections for courses in Machine Learning, Robot Learning, Data Structures & Functional Programming and Analysis of Algorithms to assist CS students (400+ students)

Cornell Cup Robotics

C1C0 CS Systems Lead/Path Planning Lead & Software Engineer

Ithaca, NY

Sept. 2021 – Dec. 2022

- Led a group of 15+ Software Engineers, facilitating seamless integration of chatbot with facial recognition and path planning
- Spearheaded a team of engineers in path planning for a semi-autonomous lab assistant robot leveraging A* search
- Integrated indoor GPS and LiDAR sensors through a server-client architecture, unifying path planning and locomotion

Northwestern Mutual

Software Engineering Intern

Milwaukee, WI

June 2022 – Aug. 2022

- Worked in an Agile environment to develop components of the content factory pipeline in order to modernize document handling in the life insurance market with e-delivery across all 50 states
- Migrated test cases from 100+ repositories into a Zero Touch Quality Assurance automated testing infrastructure

SUNY Polytechnic Institute

Neuromorphic Computing Research Intern

Albany, NY

June 2021 – August 2021

- Produced an image recognition demo application of In-Memory Vector-Matrix Multiplication achieving over 95% accuracy, inspired by AprilTags
- Researched resistive memory devices and their applications in high speed non-Von Neumann technologies
- Designed and built a GUI in Python to optimize microcontroller-to-memristor communication

PROJECTS

Munchkey! | Java, libgdx, box2d

- Led architectural designs and game development for a team of 4 programmers and 6 designers using libgdx and box2d
- Won audience vote for best desktop game in the GDIAC 2023 showcase

Show Tracker App | Python, Flask, SQL, Heroku, Docker, Postman

- Collaborated with a team of 5 developers to design a backend infrastructure with a relational database and deploy a RESTful API with 10+ HTTP routes for a show-tracking iOS application
- Received an honorable mention for Best Backend out of 25 teams in the Cornell AppDev Hackathon

Reddit Analysis Tool (NLP/ML) | OCaml, OUnit2, OWL

- Developed a system to analyze subreddits on the Reddit platform, generating text-based prediction algorithms and information using Natural Language Processing and Machine Learning techniques
- Made Reddit API requests to retrieve necessary subreddit information for analysis, including text and upvotes